
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

Anonymous authors
Anonymous submission

ABSTRACT

A substantial body of work has studied the computational complexity of deriving two fundamental families of explanations for machine-learning predictions: (i) *sufficient reasons*, which are subsets of input features that, once fixed, determine a prediction, and (ii) *contrastive reasons*, which are subsets of input features that, once modified, alter a prediction. These explanations have hitherto been analyzed in isolation across disparate contexts — non-probabilistic versus probabilistic formulations, and local versus global settings. We introduce a single unifying framework that subsumes all of these variants, establishing that each can be characterized precisely as the minimization of one common probabilistic value function. Within this framework, we prove that the complexity of every such computation is governed by three structural properties of the value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — three cornerstone properties of *combinatorial optimization*. This unifying lens exposes a sharp and counterintuitive dichotomy between the local and global regimes. Whereas the *local* value functions possess none of these properties — neither monotonicity nor submodularity/supermodularity — we prove that their *global* counterparts possess all of them. We leverage this dichotomy to establish a series of new polynomial-time results for computing explanations with provable guarantees throughout the global setting, spanning the full interpretability spectrum of model classes: neural networks, decision trees, and tree ensembles. Conversely, we prove that even drastically simplified instances of these same explanations are NP-hard to compute in the corresponding local setting, delineating a precise tractability frontier between the two regimes.

1 INTRODUCTION

Despite substantial progress in methods for explaining the decisions of machine-learning (ML) models, a recurring and sobering finding across the literature is that many of the most desirable explanation types — precisely those equipped with provable guarantees — are computationally hard to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), with this hardness intensifying for the complex, highly non-linear models where reliable explanations are needed most (Barceló et al., 2020; Adolfi et al., 2025; 2024). Consequently, the *computational complexity* of obtaining explanations has emerged as a central theoretical concern, and a growing body of work has sought to chart, for each model class, precisely which explanation types can be computed efficiently and which remain intractable (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Within the study of explanation complexity, two explanation types have received the most sustained attention, and they anchor our analysis: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features S that, once fixed to specific values, pins down the model’s prediction regardless of any assignment to the complementary set $\bar{S}$. A *contrastive reason*, dually, is a subset of input features $\bar{S}$ whose modification suffices to flip the model’s prediction.

Unlike additive attribution methods, which distribute importance scores across features yet are frequently difficult for humans to interpret (Kumar et al., 2020) and offer limited actionability (Bilodeau et al., 2024), sufficient and contrastive reasons furnish *discrete, condition-like explanations* that answer directly the two questions practitioners actually pose: “what is enough to justify this prediction?” and “what must change to flip it?”. This discrete, human-aligned character explains their central role across a broad swath of classical XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), their demonstrated superiority over additive models at improving human predictive accuracy (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and their established utility for downstream tasks including bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A further well-established principle holds that *smaller* sufficient and contrastive explanations are more interpretable, elevating *minimality* to a central guarantee of interest in its own right (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) conducted one of the earliest complexity-theoretic studies of sufficient and contrastive reasons, and their results already reveal how sharply tractability depends on both the model class and the explanation type. They established that computing a minimum-cardinality sufficient reason is NP-hard for decision trees and escalates to Σ_2^P -hard for neural networks; for contrastive reasons, the minimum-size explanation is computable in polynomial time for decision trees yet becomes NP-hard for neural networks. Analogous hardness results were subsequently extended to tree ensembles (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. The classical definitions of sufficient and contrastive reasons are widely criticized for their rigidity: because they demand that a guarantee hold in an absolute sense over the entire domain, they routinely yield explanations that are excessively large and, in practice, uninformative (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To overcome this rigidity, the field has converged on a strictly more general, *probabilistic* formulation of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023), in which one seeks subsets of input features that determine or alter a prediction with probability exceeding a prescribed threshold δ .

Wäldchen et al. (2021) initiated the complexity-theoretic study of probabilistic sufficient reasons, proving that the problem is NPP-hard already for CNF classifiers, with this hardness propagating to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). The pivotal theoretical insight of the probabilistic setting is the *failure of monotonicity* in the underlying probability function (Arenas et al., 2022; Izza et al., 2023) — a failure that renders even *subset-minimal* explanations computationally hard. Strikingly, (Arenas et al., 2022) show that computing a subset-minimal probabilistic sufficient reason is NP-hard even for decision trees, in sharp contrast to the non-probabilistic case, where the same task admits a polynomial-time algorithm (Huang et al., 2021; Barceló et al., 2020). Non-monotonicity thus stands out as the precise structural obstruction separating tractable from intractable explanation problems — a theme our framework brings to the fore.

From local to global explanations. More recently, Bassan et al. (2024) extended the complexity analysis from the *local* (non-probabilistic) setting — where an explanation is tied to a single prediction — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons valid across the entire domain. Yet, as with other non-probabilistic formulations (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), the resulting criteria are exceedingly stringent — arguably more so globally than locally: any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), so the explanation typically encompasses nearly all input features and thereby forfeits its informativeness. This observation motivates the central question of our work: does the probabilistic relaxation, which tames the strictness of local explanations, reshape the global landscape as well?

OUR CONTRIBUTIONS

1. **A unifying framework.** We unify the previously fragmented landscape of explanation-computation problems — sufficient and contrastive, probabilistic and non-probabilistic, local and global — into a single framework, casting every variant as the minimization of one common value function. We then isolate the three structural properties of this value

function that dictate the complexity of the task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.

2. **A local–global structural dichotomy.** We prove that these properties behave in *diametrically opposed* ways across the local and global regimes. While the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, we establish the surprising result that their *global* counterparts are *monotonic non-decreasing*. We further uncover structure unique to the global regime: the global sufficient value function is *supermodular* and the global contrastive value function is *submodular* — whereas locally neither property holds. This dichotomy is the analytical engine driving all of our tractability results.
3. **Polynomial-time subset-minimal global explanations.** Exploiting this structure, we prove that global explanations with provable guarantees are efficiently computable precisely where their local counterparts are intractable, and we establish this across three model classes spanning the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. Concretely, whereas computing a subset-minimal local probabilistic sufficient/contrastive reason is NP-hard even for decision trees (Arenas et al., 2022), we show its global counterpart is computable in polynomial time. We then extend this positive result to *any* model admitting polynomial-time inference — including complex black-box models such as neural networks and tree ensembles — under empirical distributions: the subset-minimal global sufficient/contrastive explanation is obtainable in polynomial time, while the local version remains NP-hard.
4. **Constant-factor approximations for cardinally minimal global explanations.** We establish a still stronger separation by exploiting the submodularity and supermodularity that hold globally but fail locally. For the harder task of computing *cardinally minimal* global explanations, we obtain provable *constant-factor* approximation guarantees — even for neural networks and tree ensembles — whenever the empirical distribution is fixed. In sharp contrast, we prove strong inapproximability for the local setting: no *bounded* approximation is achievable, even under drastically simplified assumptions.

Owing to space constraints, the main text presents our principal theorems and corollaries, with complete proofs deferred to the appendix.

2 PRELIMINARIES

Setting. We consider an input space of dimension $n \in \mathbb{N}$, with input vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$, where each coordinate $\mathbf{x}_i$ ranges over a domain $\mathcal{X}_i$ that may be discrete or continuous. The full input feature space is thus the product $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$, and we study classification models $f : \mathbb{F} \rightarrow [c]$ over $c \in \mathbb{N}$ classes. Explanations are evaluated with respect to a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$ over the input space. Since accurately approximating $\mathcal{D}$ is computationally infeasible in many settings, a standard and practical alternative is to operate over a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$ that serves as a proxy for the “true” distribution — for instance, a sampled subset of the available training data.

The explanations we study are either local or global. A *local* explanation targets a specific prediction $\mathbf{x} \in \mathbb{F}$, articulating why the model returned $f(\mathbf{x})$ on that instance. A *global* explanation, by contrast, abstracts away from any single input and characterizes the general decision-making logic governing the behavior of f across the input space as a whole.

Models. Although many of our results are fully general — notably the intrinsic properties of the value functions — we complement them with model-specific complexity results for widely deployed ML models. To span the interpretability spectrum in its entirety, we analyze model classes ranging from those regarded as “black-box” to those regarded as “interpretable”: (i) decision trees, (ii) neural networks (encompassing every architecture at least as expressive as a feed-forward ReLU network), and (iii) tree ensembles, covering both majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). Formal definitions are provided in the Appendix.

Distributions. In probabilistic explanation settings the complexity depends critically on the input distribution $\mathcal{D}$, and we therefore delineate three distribution classes that together cover the assumptions used throughout our analysis: (i) *general distributions*, which impose no assumption whatsoever on $\mathcal{D}$ and thus encompass all possible distributions; these underpin our universally valid structural

results; (ii) *empirical distributions*, comprising all distributions induced by a finite dataset $\mathbf{D}$ — the workhorse assumption of practical XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which posit mutual independence among the features of $\mathcal{D}$ — another assumption in widespread use (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Crucially, empirical distributions do *not* entail feature independence; on the contrary, they can encode the rich dependency structure extracted from finite datasets (Van den Broeck et al., 2022), which is precisely what makes our empirical-distribution tractability results applicable to realistic modeling regimes. Complete formal definitions are deferred to the Appendix.

3 FORMS OF REASONS

This section formalizes the explanation types studied throughout the paper. We begin with the strict non-probabilistic definitions of (local and global) sufficient and contrastive reasons, and then generalize them to their more flexible probabilistic counterparts, of which the non-probabilistic definitions arise as the limiting case.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In *feature selection*, one commonly identifies the subset of features responsible for a model’s decision, and the *sufficiency* criterion — adopted by a range of established explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022) — makes this notion precise. Sufficiency admits both a local instantiation, anchored to a single prediction, and a global one, quantified over the entire input domain. Following standard conventions, a *local sufficient reason* is a subset of features $S \subseteq \{1, \dots, n\}$ such that fixing the features in S to their values in $\mathbf{x}$ preserves the prediction $f(\mathbf{x})$ for every assignment to the complementary features $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the vector whose features in S are inherited from $\mathbf{x}$ and whose features in $\bar{S}$ are inherited from $\mathbf{z}$. Local sufficient reasons are intimately connected to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek an alternative input $\mathbf{z}'$ that preserves the prediction (i.e., $f(\mathbf{z}') = f(\mathbf{x})$) while remaining as “close” as possible to the original point: instantiating $\mathbf{z}'$ as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and measuring proximity by Hamming distance renders the two notions identical.

A *global sufficient reason* (Bassan et al., 2024) strengthens this to a subset of input features $S \subseteq \{1, \dots, n\}$ that constitutes a local sufficient reason simultaneously for *every* input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A complementary and equally prevalent paradigm explains a prediction by pinpointing subsets of input features whose *modification* alters it (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024), thereby identifying the minimal change required to overturn the decision. Formally, a subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are, in turn, tightly linked to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment $\mathbf{x}'$ that flips the prediction, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$; under the Hamming metric the two notions again coincide.

Mirroring the global sufficiency criterion, one may ask for a subset of input features whose alteration flips the prediction for *all* inputs in the domain of interest. This global contrastive notion connects directly to techniques for detecting bias (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and to *group* counterfactual methods that seek counterfactuals across many instances at once (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* for f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As motivated in the introduction, the non-probabilistic definitions impose conditions too stringent for practical use, which is why the field has embraced strictly more general *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021): rather than requiring a guarantee to hold absolutely, they require it to hold with probability at least $\delta \in [0, 1]$. Setting $\delta = 1$ recovers the original non-probabilistic definitions exactly, so the probabilistic formulation is a genuine generalization rather than a departure.

Probabilistic sufficient reasons. A subset $S \subseteq \{1, \dots, n\}$ is a *local δ -sufficient reason* for $\langle f, \mathbf{x} \rangle$ if fixing the features in S to their values in $\mathbf{x}$ and sampling the remaining features from $\mathcal{D}$ preserves the classification with probability at least δ ; that is:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ indicates that the features in S of $\mathbf{z}$ are fixed to their values in $\mathbf{x}$. Following the standard construction of global explanations by averaging over the input domain, we define a subset $S \subseteq \{1, \dots, n\}$ to be a *global δ -sufficient reason* for $\langle f \rangle$ when the expectation of the local sufficiency probability, taken over inputs drawn from $\mathcal{D}$, is at least δ ; that is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. Dually, a *local δ -contrastive reason* is a subset of input features whose variation flips the prediction with sufficient probability. In contrast to the sufficient case, here the *complementary* features $\bar{S}$ are fixed to their values in $\mathbf{x}$ while the features in S are allowed to vary; S qualifies when the prediction differs from $f(\mathbf{x})$ with probability at least δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

In the global setting, and in exact analogy with global sufficient reasons, a subset S is a *global δ -contrastive reason* when the expectation of the local contrastive probability over inputs drawn from $\mathcal{D}$ is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

As noted in the introduction, across every explanation form introduced so far — sufficient or contrastive, local or global — the literature converges on the principle that *smaller* explanations are more meaningful, which elevates *minimality* to a provable guarantee of primary interest. We study two central and complementary notions of minimality, both applicable uniformly to all of our explanation types:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the *smallest explanation cardinality* $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal S is subset-minimal, whereas the converse fails (Appendix B.3 gives an explicit example). We adopt the terms *subset* and *cardinally* minimal, in preference to local and global minima, to forestall any confusion with local versus global explanations (input-level versus domain-level reasoning). Both notions specialize uniformly to *every* explanation form we analyze: a cardinally minimal local probabilistic δ -sufficient reason, for instance, is the qualifying S of smallest $|S|$, while a subset-minimal one is any qualifying S admitting no qualifying proper subset $S' \subsetneq S$.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

A central observation underpinning our framework is that all of the computational problems introduced above — local or global, sufficient or contrastive, probabilistic or non-probabilistic — reduce to a single template: find a minimum-size subset S with $v(S) \geq \delta$, where the non-probabilistic variants correspond precisely to $\delta := 1$. To make this uniform, we introduce notation for the four value functions. Let v_{suff}^ℓ denote the local sufficiency probability of equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and let v_{suff}^g denote its global counterpart (equation 6); analogously, let v_{con}^ℓ and v_{con}^g denote the local and global contrastive probabilities (equations 7 and 73, respectively). With this notation in place, the unified task of computing a cardinally minimal local/global sufficient/contrastive δ -reason is stated precisely as follows:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

For the relaxed objective of computing a subset-minimal rather than cardinally minimal local or global sufficient/contrastive reason, the corresponding task is defined as follows:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subseteq S$ it holds that $v(S') < \delta$.

Properties of v that govern the complexity. We now isolate the structural properties of the value function that, as our subsequent analysis establishes, determine the complexity of computing explanations. The first is *non-decreasing monotonicity*, which guarantees that the marginal contribution $v(S \cup \{i\}) - v(S)$ of any feature is always non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The remaining two properties, *supermodularity* and *submodularity*, concern how the marginal contribution $v(S \cup \{i\}) - v(S)$ evolves as the set S grows. In the supermodular case this marginal contribution is itself non-decreasing in S (features become more valuable the more context is fixed), whereas in the dual submodular case it is non-increasing (features exhibit diminishing returns). Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

In the local non-probabilistic setting, prior work shows that subset-minimal sufficient or contrastive reasons are computable precisely because the value function is monotone — but this monotonicity holds only in the restricted regime $\delta = 1$, which severely curtails practical flexibility. The natural remedy of moving to probabilistic guarantees for arbitrary δ was shown to backfire: monotonicity collapses in the local probabilistic setting, and the computation of explanations becomes correspondingly harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b). This is exactly the obstruction our global analysis overcomes.

The global non-probabilistic setting is even more extreme: Bassan et al. (2024) established a stringent *uniqueness* property — there exists exactly *one* subset-minimal explanation — but only under $\delta = 1$, an assumption especially restrictive globally, where the sufficiency condition must hold for *all* inputs. Indeed, Bassan et al. (2024) shows that this unique minimal subset coincides with the set of all non-strictly-redundant features, which in practice may be the entire feature set. We prove that in the

general global *probabilistic* setting — for arbitrary δ — this uniqueness property emphatically fails: the number of subset-minimal global sufficient reasons can be *exponential*.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Although uniqueness thus survives only in the degenerate case $\delta = 1$, the property that truly matters for computation — *monotonicity* — holds for the global value function across *all* values of δ , and for both the sufficient (v_{suff}^g) and contrastive (v_{con}^g) value functions. This is a striking reversal of the local picture, where the analogous functions (v_{suff}^l and v_{con}^l) are provably non-monotone:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^l and v_{suff}^l are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond monotonicity — which holds globally yet fails locally — the global setting possesses further structure with no local analogue, namely submodularity and supermodularity. Under the standard assumption of feature independence, we prove that the global sufficient value function v_{suff}^g is supermodular, whereas its local counterpart v_{suff}^l fails supermodularity even under the far more restrictive uniform (hence independent) distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^l is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

For the second family of explanations — global probabilistic *contrastive* reasons — the corresponding value function exhibits the dual property: it is not supermodular but *submodular*. This complementarity between the sufficient and contrastive cases is again in sharp contrast to the local setting, where the value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

We now analyze the complexity of computing *subset-minimal* explanations (local/global, sufficient/contrastive) across the model classes under study. The decisive property throughout is *monotonicity*. As established in the previous section, monotonicity holds for both global value functions, v_{con}^g and v_{suff}^g , but fails for both local ones, v_{con}^l and v_{suff}^l . This single distinction is what makes a greedy procedure converge to a subset-minimal explanation globally while failing to do so locally, and it yields the central finding of this subsection: computing subset-minimal global explanations is tractable (polynomial-time solvable), whereas computing their local counterparts is hard. We begin with the following generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

```

▷ S is a (subset minimal?) δ -explanation

Algorithm 1 computes a candidate subset-minimal δ -explanation via a top-down greedy strategy. It initializes S to the full feature set $\{1, \dots, n\}$ and, at each iteration, removes the feature j that inflicts

the least damage on the objective — i.e., the j maximizing $v(S \setminus \{j\})$ — continuing while some single-feature removal still keeps the value at or above δ (i.e., while $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$). Upon termination it returns S .

Whether Algorithm 1 actually returns a subset-minimal explanation hinges entirely on the *monotonicity* of v . The algorithm halts once removing any single feature from S drives $v(S \setminus \{i\})$ below δ ; monotonicity upgrades this local stopping condition into the far stronger global guarantee that $v(S \setminus S') < \delta$ for *every* nonempty S' , which is exactly subset-minimality. Combining this with the monotonicity of the global value functions established in the previous sections yields the following proposition:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^l and v_{suff}^l does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building on this convergence guarantee, we now establish new complexity results for computing the various forms of subset-minimal explanations in our framework, model class by model class.

Decision trees. Consider first the most benign, ostensibly “interpretable” regime: f is a decision tree and $\mathcal{D}$ is an independent distribution. Even here a strict separation emerges. Arenas et al. (2022) proved the intractability result that, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees under independent distributions — indeed, even under the uniform distribution. We prove that the very same problem becomes tractable in the global regime: a subset-minimal global δ -sufficient reason for decision trees is computable in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. The same tractability guarantee for global explanations extends to *orthogonal DNFs* (Crama & Hammer, 2011) — disjunctive normal form formulas whose terms are pairwise mutually exclusive. This extension is significant because orthogonal DNFs strictly *generalize* decision trees while retaining the structural properties that make the value function efficiently computable, so our guarantee reaches beyond tree-structured models. The full argument appears in Appendix N.

Neural networks, tree ensembles, and other complex models. We now push these results well beyond decision trees. We show that whenever $\mathcal{D}$ is an empirical distribution and f admits polynomial-time inference — a mild condition satisfied by essentially every model deployed in practice — computing a subset-minimal global δ -sufficient or contrastive reason is achievable in polynomial time, regardless of the model’s internal complexity.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

This positive result, which holds for *any* model under an empirical distribution, sets up a further sharpening of the local–global separation. For sufficiently expressive models, computing subset-minimal *local* explanations remains intractable even when the distribution is empirical. We establish this hardness for both neural networks and tree ensembles, obtaining the following strict complexity separation:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now turn from subset-minimal reasons to the more demanding task of computing a *cardinally minimal* δ -sufficient/contrastive reason. In the global setting, the interplay between monotonicity

and submodularity/supermodularity yields *provable approximation* guarantees for this task; in the local setting, by contrast, we prove that even bounded approximation is unattainable. This widens the local–global tractability gap from an exact-computation phenomenon into an approximation-theoretic one.

A unified greedy approximation algorithm. For a monotone non-decreasing submodular value function, computing a cardinally minimal explanation is an instance of the *submodular set cover* problem (Wolsey, 1982); for a monotone *supermodular* function, it becomes the harder non-submodular variant (Shi et al., 2021). Both problems are classical objects of study, prized for their strong approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). The canonical solver for submodular set cover is a greedy algorithm, which we present next (Algorithm 2); it is the common engine behind our approximations of both cardinally minimal sufficient and contrastive δ -reasons, whose respective guarantees we analyze in turn.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation
```

Algorithm 2 is the bottom-up dual of Algorithm 1: it starts from the empty set and greedily augments S , at each step adding the feature that maximizes $v(S \cup \{i\})$, until the value first reaches the threshold δ .

Cardinally minimal contrastive reasons. We treat global contrastive reasons first, since here both monotonicity and submodularity hold, casting the task exactly as *submodular set cover* and letting us invoke the classical guarantee of Wolsey (1982) through Algorithm 2. For integer-valued objectives the algorithm attains a Harmonic-based factor, and in general an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ of fixed sample size this ratio collapses to a *constant* — growing only *logarithmically* in the sample size — and thus constitutes a strong guarantee. The *local* setting could hardly be more different: even for a *single* sample point, no *bounded* approximation exists, marking a categorical gap between the two regimes.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We further establish a matching *lower bound* for Theorem 3 in the case of an empirical distribution without any independence assumption, certifying that the guarantee is essentially tight (see Appendix M).

Cardinally minimal sufficient reasons. Whereas cardinally minimal global *contrastive* reasons map onto submodular set cover and thereby inherit strong approximations, cardinally minimal global *sufficient* reasons correspond to the supermodular variant, which is inherently harder to approximate — yet still admits guarantees once the function’s curvature is bounded (Shi et al., 2021). Recall the *total curvature* of $v : 2^n \rightarrow \mathbb{R}$, defined as $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Invoking Shi et al. (2021), we show that Algorithm 2 attains an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

As in the contrastive case, under a fixed empirical distribution this approximation is constant. The sufficient case is quantitatively weaker — it carries the additional $\frac{1}{1-k^f}$ factor absent from the tighter $\mathcal{O}(\ln(|D|))$ contrastive bound — but it remains a genuine constant-factor approximation. And once again the local variant stands in stark opposition: it is inapproximable, admitting no *bounded* approximation even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k\tau} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Taken together, these results reinforce and extend Subsection 7.1: having shown that subset-minimal global explanations are tractable where their local counterparts are hard, we have now shown that even the harder cardinally minimal global explanations are efficiently approximable, whereas their local counterparts are inapproximable. The local–global divide thus persists across both minimality notions and both the exact and approximate regimes.

8 LIMITATIONS AND FUTURE WORK

Our most fundamental findings — the structural properties of the value functions — hold in full generality, and we instantiate them to obtain concrete complexity results for specific model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions within our framework. Settings beyond those instantiated here remain open, and we regard this breadth of unexplored ground not as a shortcoming but as an indication of the framework’s reach: the general structural results and the new tractability and intractability separations they yield furnish a principled foundation on which a far wider range of explainability scenarios can be analyzed.

Our study opens several concrete theoretical directions:

1. First, the tractable global-explainability results established here may admit further tightening for structurally simple model classes such as decision trees, linear models, and other inherently tractable models. Although local explanations for these models are frequently intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the strong structural properties of the global explanation forms we study already yield markedly improved guarantees that we conjecture can be sharpened further; tightening the approximation factors under empirical distributions is a particularly promising avenue.
2. Second, when the relevant expectations cannot be computed *exactly*, it is natural to ask whether they can be approximated — for instance, via Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes such as DNF formulas (Meel et al., 2019) (and hence tree ensembles), or via Monte Carlo estimators of the kind used in (Subercaseaux et al., 2025). The central challenge is to characterize the tolerance under which the minimality guarantees provably survive approximation error.
3. Finally, we expect that alternative notions of feature importance — e.g., loss measures such as KL divergence (Conmy et al., 2023) — exhibit the same local-to-global emergence of monotonicity, submodularity, or supermodularity, which would extend efficient feature selection to those metrics as well.

9 CONCLUSION

We have introduced a unified framework for analyzing diverse explanation types and used it to expose a stark structural contrast between local and global sufficient and contrastive reasons. Whereas the *local* variants possess none of the properties central to combinatorial optimization — monotonicity, submodularity, or supermodularity — we prove that their *global* counterparts possess exactly these properties: (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These structural results are the foundation for a series of complexity separations establishing that global explanations with provable guarantees are efficiently computable — even for complex model classes such as neural networks — while their local counterparts remain NP-hard even in drastically simplified settings. In uncovering these foundational properties and charting the precise frontier between tractability and intractability, our work supplies both the conceptual framework and the technical machinery for a broad program of future research on the computation of formal explanations.

REFERENCES

- Federico Adolphi, Martina G Vilas, and Todd Wareham. Complexity-Theoretic Limits on the Promises of Artificial Neural Network Reverse-Engineering. In *Proc. 46th Annual Meeting of the Cognitive Science Society (CogSci)*, 2024.
- Federico Adolphi, Martina Vilas, and Todd Wareham. The Computational Complexity of Circuit Discovery for Inner Interpretability. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025.
- Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and Irina Trubitsyna. Even-If Explanations: Formal Foundations, Priorities and Complexity. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 15347–15355, 2025.
- Ibrahim M Almanjahie, Zouaoui Chikr Elmezouar, Ali Laksaci, and Mustapha Rachdi. KNN Local Linear Estimation of the Conditional Cumulative Distribution Function: Dependent Functional Data Case. *Comptes Rendus. Mathématique*, 356(10):1036–1039, 2018.
- David Alvarez Melis and Tommi Jaakkola. Towards Robust Interpretability with Self-Explaining Neural Networks. In *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.
- Leila Amgoud and Martin Cooper. Axiomatic Foundations of Counterfactual Explanations. 2026. Technical Report. <https://arxiv.org/abs/2602.04028>.
- Guy Amir, Shahaf Bassan, and Guy Katz. Hard to Explain: On the Computational Hardness of In-Distribution Model Interpretation. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024.
- Alaa Anani, Tobias Lorenz, Mario Fritz, and Bernt Schiele. Pixel-level Certified Explanations via Randomized Smoothing. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.
- Marcelo Arenas, D. Baez, P. Barceló, J. Pérez, and B. Subercaseaux. Foundations of Symbolic Languages for Model Interpretability. In *Proc. 34th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 11690–11701, 2021.
- Marcelo Arenas, P. Barceló, M. Romero, and B. Subercaseaux. On Computing Probabilistic Explanations for Decision Trees. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 28695–28707, 2022.
- Marcelo Arenas, Pablo Barceló, Leopoldo Bertossi, and Mikaël Monet. On the Complexity of SHAP-Score-Based Explanations: Tractability via Knowledge Compilation and Non-Approximability Results. *Journal of Machine Learning Research (JMLR)*, 24(63):1–58, 2023.
- Gilles Audemard, S. Bellart, L. Bounia, F. Koriche, J. Lagniez, and P. Marquis. On the Computational Intelligibility of Boolean Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 74–86, 2021.
- Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre Marquis. On Preferred Abductive Explanations for Decision Trees and Random Forests. In *Proc. 31st Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 643–650, 2022a.
- Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre Marquis. Trading Complexity for Sparsity in Random Forest Explanations. In *Proc. 36th AAAI Conf. on Artificial Intelligence*, number 5, pp. 5461–5469, 2022b.
- Gilles Audemard, Jean-Marie Lagniez, Pierre Marquis, and Nicolas Szczepanski. Computing Abductive Explanations for Boosted Trees. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 4699–4711, 2023.
- Steve Azzolin, Sagar Malhorta, Andrea Passerini, and Stefano Teso. Beyond Topological Self-Explainable GNNs: A Formal Explainability Perspective. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.

-
- Esma Balkir, Isar Nejadgholi, Kathleen C Fraser, and Svetlana Kiritchenko. Necessity and Sufficiency for Explaining Text Classifiers: A Case Study in Hate Speech Detection. In *Proc. Conf. of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 2672–2686, 2022.
- Pablo Barceló, M. Monet, J. Pérez, and B. Subercaseaux. Model Interpretability through the Lens of Computational Complexity. In *Proc. 33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 15487–15498, 2020.
- Pablo Barceló, Alexander Kozachinskiy, Miguel Romero, Bernardo Subercaseaux, and José Verschae. Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations. In *Proc. 3rd ACM on Management of Data*, pp. 1–26, 2025.
- Shahaf Bassan and G. Katz. Towards Formal XAI: Formally Approximate Minimal Explanations of Neural Networks. In *Proc. 29th Int. Conf. on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, pp. 187–207, 2023.
- Shahaf Bassan, Guy Amir, Davide Corsi, Idan Refaeli, and Guy Katz. Formally Explaining Neural Networks Within Reactive Systems. In *Proc. 23rd Int. Conf. on Formal Methods in Computer-Aided Design (FMCAD)*, pp. 1–13, 2023.
- Shahaf Bassan, Guy Amir, and Guy Katz. Local vs. Global Interpretability: A Computational Complexity Perspective. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.
- Shahaf Bassan, Guy Amir, Meirav Zehavi, and Guy Katz. What makes an Ensemble (Un) Interpretable? In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025a.
- Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Matthias Althoff, and Guy Katz. Explaining, Fast and Slow: Abstraction and Refinement of Provable Explanations. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025b.
- Shahaf Bassan, Ron Eliav, and Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural Networks with Concise Sufficient Reasons. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025c.
- Shahaf Bassan, Shlomit Gur, Sergey Zeltyn, Konstantinos Mavrogiorgos, Ron Eliav, and Dimosthenis Kyriazis. Self-Explaining Neural Networks for Business Process Monitoring. 2025d. Technical Report. <https://arxiv.org/abs/2503.18067>.
- Shahaf Bassan, Michal Moshkovitz, and Guy Katz. Additive Models Explained: A Computational Complexity Approach. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025e.
- Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Volkan Şahin, Jan Kretinsky, Matthias Althoff, and Guy Katz. Provably Explaining Neural Additive Models. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*, 2026.
- Robi Bhattacharjee and Ulrike Luxburg. Auditing Local Explanations is Hard. In *Proc. 38th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 18593–18632, 2024.
- Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility Theorems for Feature Attribution. *Proc. of the National Academy of Sciences (PNAS)*, 121(2):e2304406120, 2024.
- Guy Blanc, Jane Lange, and Li-Yang Tan. Provably Efficient, Succinct, and Precise Explanations. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 6129–6141, 2021.
- Guy Blanc, Caleb Koch, Jane Lange, and Li-Yang Tan. A Query-Optimal Algorithm for Finding Counterfactuals. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 2075–2090, 2022.
- Ryma Boumazouza, Fahima Cheikh-Alili, Bertrand Mazure, and Karim Tabia. ASTERYX: A model-Agnostic SaT-basEd appRoach for SYmbolic and Score-Based EXplanations. In *Proc. 30th ACM Int. Conf. on Information & Knowledge Management (CIKM)*, pp. 120–129, 2021.

-
- Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, Shahaf Bassan, and Guy Katz. FAME: Formal Abstract Minimal Explanation for Neural Networks. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*, 2026.
- Louenas Bounia. Enhancing the Intelligibility of Boolean Decision Trees with Concise and Reliable Probabilistic Explanations. In *Proc. 20th Int. Conf. on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU)*, pp. 205–218, 2024.
- Louenas Bounia. Using Submodular Optimization to Approximate Minimum-Size Abductive Path Explanations for Tree-Based Models. In *Proc. 41st Conf. on Uncertainty in Artificial Intelligence (UAI)*, 2025.
- Louenas Bounia and Frederic Koriche. Approximating Probabilistic Explanations via Supermodular Minimization. In *Proc. 39th Conf. on Uncertainty in Artificial Intelligence (UAI)*, pp. 216–225, 2023.
- Marco Calautti, Enrico Malizia, and Cristian Molinaro. On the Complexity of Global Necessary Reasons to Explain Classification. 2025. Technical Report. <https://arXiv.org/abs/2501.06766>.
- Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical Optimization Modelling for Group Counterfactual Explanations. *European Journal of Operational Research (EJOR)*, 319(2):399–412, 2024.
- Brandon Carter, Jonas Mueller, Siddhartha Jain, and David Gifford. What Made You Do This? Understanding Black-Box Decisions with Sufficient Input Subsets. In *Proc. 22nd Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 567–576, 2019.
- Brandon Carter, Siddhartha Jain, Jonas W Mueller, and David Gifford. Overinterpretation Reveals Image Classification Model Pathologies. In *Proc. 35th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 15395–15407, 2021.
- Wenjing Chen and Victoria Crawford. Bicriteria Approximation Algorithms for the Submodular Cover Problem. In *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 72705–72716, 2023.
- Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards Automated Circuit Discovery for Mechanistic Interpretability. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 16318–16352, 2023.
- Yves Crama and Peter L Hammer. *Boolean Functions: Theory, Algorithms, and Applications*. Cambridge University Press, 2011.
- Adnan Darwiche. Logic for Explainable AI. In *Proc. 38th ACM/IEEE Symposium on Logic in Computer Science (LICS)*, pp. 1–11, 2023.
- Adnan Darwiche and Auguste Hirth. On the Reasons Behind Decisions. In *Proc. 24th European Conf. on Artificial Intelligence (ECAI)*, pp. 712–720, 2020.
- Adnan Darwiche and Chunxi Ji. On the Computation of Necessary and Sufficient Explanations. In *Proc. 36th AAAI Conf. on Artificial Intelligence*, pp. 5582–5591, 2022.
- Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for Evaluating Faithfulness of Local Explanations. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 4794–4815, 2022.
- Ian Davidson, Nicolás Kennedy, and SS Ravi. CXAD: Contrastive Explanations for Anomaly Detection: Algorithms, Complexity Results and Experiments. *Transactions on Machine Learning Research (TMLR)*, 2025.
- Alessandro De Palma, Greta Dolcetti, and Caterina Urban. Faster Verified Explanations for Neural Networks. 2025. Technical Report. <https://arXiv.org/abs/2512.00164>.
- Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam, and Payel Das. Explanations Based on the Missing: Towards Contrastive Explanations with Pertinent Negatives. *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

- 702 Hidde Fokkema, Rianne De Heide, and Tim Van Erven. Attribution-Based Explanations that Provide
703 Recourse Cannot be Robust. *Journal of Machine Learning Research (JMLR)*, 24(360):1–37, 2023.
704
- 705 Hidde Fokkema, Damien Garreau, and Tim van Erven. The Risks of Recourse in Binary Classification.
706 In *Proc. 27th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 550–558, 2024.
- 707 Tobias Geibinger, Reijo Jaakkola, Antti Kuusisto, Xinghan Liu, and Miikka Vilander. Why This and
708 Not That? A Logic-based Framework for Contrastive Explanations. In *Proc. 19th European Conf.
709 on Logics in Artificial Intelligence (JELIA)*, pp. 45–60. Springer, 2025.
- 710 Joshua Größer and Ostap Okhrin. Copulae: An Overview and Recent Developments. *Wiley Interdis-*
711 *ciplinary Reviews: Computational Statistics*, 14(3):e1557, 2022.
712
- 713 Riccardo Guidotti. Counterfactual Explanations and How to Find Them: Literature Review and
714 Benchmarking. *Data Mining and Knowledge Discovery (DMKD)*, 38(5):2770–2824, 2024.
- 715 Hyukjun Gweon, Matthias Schonlau, and Stefan H Steiner. The K Conditional Nearest Neighbor
716 Algorithm for Classification and Class Probability Estimation. *PeerJ Computer Science*, 5:e194,
717 2019.
- 718 Itamar Hadad, Shahaf Bassan, and Guy Katz. Formal Mechanistic Interpretability: Automated Circuit
719 Discovery with Provable Guarantees. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*,
720 2026.
721
- 722 Ouns El Harzli, Bernardo Cuenca Grau, and Ian Horrocks. Cardinality-Minimal Explanations for
723 Monotonic Neural Networks. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp.
724 3677–3685, 2023.
- 725 W Keith Hastings. Monte Carlo Sampling Methods using Markov Chains and their Applications.
726 *Biometrika*, 57:97–109, 1970.
727
- 728 Rongjie Huang, Max W. Y. Lam, Jun Wang, Dan Su, Dong Yu, Yi Ren, and Zhou Zhao. FastDiff: A
729 Fast Conditional Diffusion Model for High-Quality Speech Synthesis. In *Proc. 31st Int. Joint Conf.
730 on Artificial Intelligence (IJCAI)*, pp. 4157–4163, 2022.
- 731 Xuanxiang Huang, Yacine Izza, Alexey Ignatiev, and Joao Marques-Silva. On Efficiently Explaining
732 Graph-Based Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and
733 Reasoning (KR)*, 2021.
- 734 Alexey Ignatiev, Nina Narodytska, and Joao Marques-Silva. Abduction-Based Explanations for
735 Machine Learning Models. In *Proc. 33rd AAAI Conference on Artificial Intelligence*, pp. 1511–
736 1519, 2019.
- 737 Alexey Ignatiev, Martin Cooper, Mohamed Siala, Emmanuel Hebrard, and Joao Marques-Silva.
738 Towards Formal Fairness in Machine Learning. In *Proc. 26th Int. Conf. on Principles and Practice
739 of Constraint Programming (CP)*, pp. 846–867, 2020a.
- 740 Alexey Ignatiev, Nina Narodytska, Nicholas Asher, and Joao Marques-Silva. From Contrastive to
741 Abductive Explanations and Back Again. In *Proc. Int. Conf. of the Italian Association for Artificial
742 Intelligence (AIxAI)*, pp. 335–355, 2020b.
- 743 Alexey Ignatiev, Yacine Izza, Peter J Stuckey, and Joao Marques-Silva. Using MaxSAT for Efficient
744 Explanations of Tree Ensembles. In *Proc. 36th AAAI Conference on Artificial Intelligence*, pp.
745 3776–3785, 2022.
746
- 747 Rishabh K Iyer and Jeff A Bilmes. Submodular Optimization with Submodular Cover and Submodular
748 Knapsack Constraints. *Proc. 27th Conf. on Neural Information Processing Systems (NeurIPS)*,
749 2013.
750
- 751 Yacine Izza and Joao Marques-Silva. On Explaining Random Forests with SAT. In *Proc. 30th Int.
752 Joint Conf. on Artificial Intelligence (IJCAI)*, 2021.
753
- 754 Yacine Izza, Xuanxiang Huang, Alexey Ignatiev, Nina Narodytska, Martin Cooper, and Joao Marques-
755 Silva. On Computing Probabilistic Abductive Explanations. *Int. Journal of Approximate Reasoning
(IJAR)*, 159:108939, 2023.

756 Yacine Izza, Xuanxiang Huang, Antonio Morgado, Jordi Planes, Alexey Ignatiev, and Joao Marques-
757 Silva. Distance-Restricted Explanations: Theoretical Underpinnings & Efficient implementation.
758 In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp.
759 475–486, 2024a.

760 Yacine Izza, Kuldeep S Meel, and Joao Marques-Silva. Locally-Minimal Probabilistic Explanations.
761 In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024b.

762
763 Reijo Jaakkola, Tomi Janhunen, Antti Kuusisto, Masood Feyzbakhsh Rankooh, and Miikka Vilander.
764 Explainability via Short Formulas: the Case of Propositional Logic with Implementation. *Journal*
765 *of Artificial Intelligence Research (JAIR)*, 83, 2025.

766 Alon Jacovi, Swabha Swayamdipta, Shauli Ravfogel, Yanai Elazar, Yejin Choi, and Yoav Goldberg.
767 Contrastive Explanations for Model Interpretability. In *Proc. Conf. Conference on Empirical*
768 *Methods in Natural Language Processing (EMNLP)*, pp. 1597–1611, 2021.

769
770 Helen Jin, Anton Xue, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees
771 for Feature Attributions. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*,
772 2025.

773 Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counter-
774 factual explanations to interventions. In *Proc. 4th ACM Conf. on Fairness, Accountability, and*
775 *Transparency (FAccT)*, pp. 353–362, 2021.

776
777 Eoin M Kenny and Mark T Keane. On Generating Plausible Counterfactual and Semi-Factual
778 Explanations for Deep Learning. In *Proc. 35th AAAI Conf. on Artificial Intelligence*, number 13,
779 pp. 11575–11585, 2021.

780 Alexander Kozachinskiy. Inapproximability of Sufficient Reasons for Decision Trees. 2023. Technical
781 Report. <https://arXiv.org/abs/2304.02781>.

782
783 I Elizabeth Kumar, Suresh Venkatasubramanian, Carlos Scheidegger, and Sorelle Friedler. Problems
784 with Shapley-Value-Based Explanations as Feature Importance Measures. In *Proc. 37th Int. Conf.*
785 *on Machine Learning (ICML)*, pp. 5491–5500, 2020.

786 Emanuele La Malfa, Agnieszka Zbrzezny, Rhiannon Michelmore, Nicola Paoletti, and Marta
787 Kwiatkowska. On Guaranteed Optimal Robust Explanations for NLP Models. In *Proc. Int.*
788 *Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 2658–2665, 2021.

789
790 Eduardo Sany Laber. The Computational Complexity of some Explainable Clustering Problems.
791 *Information Processing Letters*, 184:106437, 2024.

792
793 Shuai Li, Yingjie Zhang, Hongtu Zhu, Christina Wang, Hai Shu, Ziqi Chen, Zhuoran Sun, and
794 Yanfeng Yang. K-Nearest-Neighbor Local Sampling Based Conditional Independence Testing.
795 *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 23321–23344, 2023.

796 Gianluigi Lopardo, Frédéric Precioso, and Damien Garreau. A Sea of Words: An In-Depth Analysis
797 of Anchors for Text Data. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*
798 2023, 2023.

799
800 Scott M Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. *Proc. 31st*
801 *Conf. on Neural Information Processing Systems (NeurIPS)*, 2017.

802
803 Joao Marques-Silva and Alexey Ignatiev. Delivering Trustworthy AI through formal XAI. In *Proc.*
804 *36th AAAI Conf. on Artificial Intelligence*, pp. 12342–12350, 2022.

805
806 Joao Marques-Silva and Alexey Ignatiev. No Silver Bullet: Interpretable ML Models Must be
807 Explained. *Frontiers in Artificial Intelligence*, 6:1128212, 2023.

808
809 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
Explaining Naive Bayes and Other Linear Classifiers with Polynomial Time and Delay. In *Proc.*
33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS), pp. 20590–
20600, 2020.

-
- Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska. Explanations for Monotonic Classifiers. In *Proc. Int. Conf. on Machine Learning (ICML)*, pp. 7469–7479, 2021.
- Reda Marzouk and Colin De La Higuera. On the Tractability of SHAP Explanations under Markovian Distributions. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, pp. 34961–34986, 2024.
- Reda Marzouk, Shahaf Bassan, and Guy Katz. SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025a.
- Reda Marzouk, Shahaf Bassan, Guy Katz, and De la Higuera. On the Computational Tractability of the (Many) Shapley Values. In *Proc. 28th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2025b.
- Kuldeep S Meel, Aditya A Shrotri, and Moshe Y Vardi. Not all FPRASs are Equal: Demystifying FPRASs for DNF-Counting. *Constraints*, 24(3):211–233, 2019.
- Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining Machine Learning Classifiers Through Diverse Counterfactual Explanations. In *Proc. 3rd Conf. on Fairness, Accountability, and Transparency (FAccT)*, pp. 607–617, 2020.
- Vidya Muthukumar, Tejaswini Pedapati, Nalini Ratha, Prasanna Sattigeri, Chai-Wah Wu, Brian Kingsbury, Abhishek Kumar, Samuel Thomas, Aleksandra Mojsilovic, and Kush R Varshney. Understanding Unequal Gender Classification Accuracy from Face Images. 2018. Technical Report. <https://arXiv.org/abs/1812.00099>.
- Sebastian Ordyniak, Giacomo Paesani, and Stefan Szeider. The Parameterized Complexity of Finding Concise Local Explanations. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2023.
- Sebastian Ordyniak, Giacomo Paesani, Mateusz Rychlicki, and Stefan Szeider. Explaining Decisions in ML Models: A Parameterized Complexity Analysis. In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 563–573, 2024.
- Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Anchors: High-Precision Model-Agnostic Explanations. In *Proc. 32nd AAAI Conf. on Artificial Intelligence*, 2018.
- Majun Shi, Zishen Yang, and Wei Wang. Minimum Non-Submodular Cover Problem with Applications. *Applied Mathematics and Computation*, 410:126442, 2021.
- Andy Shih, Arthur Choi, and Adnan Darwiche. A Symbolic Approach to Explaining Bayesian Network Classifiers. In *Proc. 27th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 5103–5111, 2018.
- Bernardo Subercaseaux, Marcelo Arenas, and Kuldeep S Meel. Probabilistic Explanations for Linear Models. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 20655–20662, 2025.
- Guy Van den Broeck, Anton Lykov, Maximilian Schleich, and Dan Suciu. On the Tractability of SHAP Explanations. *Journal of Artificial Intelligence Research (JAIR)*, 74:851–886, 2022.
- Gal Vardi, Gilad Yehudai, and Ohad Shamir. On the Optimal Memorization Power of ReLU Neural Networks. In *Proc. 10th Int. Conf. on Learning Representations, (ICLR)*, 2022.
- Vijay V Vazirani. *Approximation Algorithms*. Springer, 2001.
- Sahil Verma, Varich Boonsanong, Minh Hoang, Keegan Hines, John Dickerson, and Chirag Shah. Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review. *ACM Computing Surveys*, 56(12):1–42, 2024.
- Stephan Wäldchen, Jan Macdonald, Sascha Hauch, and Gitta Kutyniok. The Computational Complexity of Understanding Binary Classifier Decisions. *Journal of Artificial Intelligence Research (JAIR)*, 70:351–387, 2021.
- E. Wang, P. Khosravi, and G. Van den Broeck. Probabilistic Sufficient Explanations. In *Proc. 30th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021a.

-
- 864 S. Wang, H. Zhang, K. Xu, X. Lin, S. Jana, C. Hsieh, and Z. Kolter. Beta-Crown: Efficient Bound
865 Propagation with Per-Neuron Split Constraints for Neural Network Robustness Verification. In
866 *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
867 29909–29921, 2021b.
- 868 Greta Warren, Eoin Delaney, Christophe Guéret, and Mark T Keane. Explaining Multiple Instances
869 Counterfactually: User Tests of Group-Counterfactuals for XAI. In *Proc. 32nd Int. Conf. on*
870 *Case-Based Reasoning (ICCBR)*, pp. 206–222, 2024.
- 871 Ingo Wegener. The Complexity of Symmetric Boolean Functions. *Computation Theory and Logic*,
872 pp. 433–442, 2005.
- 873 Laurence A Wolsey. An Analysis of the Greedy Algorithm for the Submodular Set Covering Problem.
874 *Combinatorica*, 2(4):385–393, 1982.
- 875 Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan
876 Refaeli, Guy Amir, Kyle Julian, Shahaf Bassan, et al. Marabou 2.0: A Versatile Formal Analyzer
877 of Neural Networks. In *Proc. 36th Int. Conf. on Computer Aided Verification (CAV)*, pp. 249–264,
878 2024a.
- 879 Min Wu, Haoze Wu, and Clark Barrett. Verix: Towards Verified Explainability of Deep Neural
880 Networks. In *Proc. 36th Int. Conf. on Advances in Neural Information Processing Systems*
881 *(NeurIPS)*, 2024b.
- 882 Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling Tabular
883 Data Using Conditional GAN. *Proc. 33rd Conf. on Neural Information Processing Systems*
884 *(NeurIPS)*, 2019.
- 885 Anton Xue, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multi-
886 plicative Smoothing. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
887 62388–62413, 2023.
- 888 Kayo Yin and Graham Neubig. Interpreting Language Models with Contrastive Explanations. In
889 *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 184–198, 2022.
- 890 Jinqiang Yu, Alexey Ignatiev, Peter Stuckey, Nina Narodytska, and Joao Marques-Silva. Eliminating
891 the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background
892 Knowledge in the Context of Formal Explanations. In *Proc. 37th AAAI Conf. on Artificial*
893 *Intelligence*, pp. 4123–4131, 2023.
- 894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917

Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work belongs to the research program of *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations endowed with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central theme of this program is the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because formal explanations are typically intractable to compute for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), much of the literature has retreated to structurally simpler model classes (Marques-Silva & Ignatiev, 2023), including decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026). Against this backdrop, our global tractability results are notable precisely because they reach the expressive models that local analyses cannot.

Complementing these complexity-theoretic investigations, a substantial body of work computes formal explanations in practice through automated reasoning: MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural-network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for deriving certified explanations of neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Further approaches mitigate the inherent computational burden via abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, a closely related line of work within formal XAI applies combinatorial and submodular optimization techniques to the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), chiefly for decision-tree classifiers. As that literature observes — and as we confirm here — the value function underlying these local variants is not submodular, so submodular-inspired greedy methods function there only as heuristics, without formal guarantees, albeit often yielding high-quality explanations in practice. Our work instead forges a rigorous connection between combinatorial and submodular optimization objectives and the *global* variants of sufficient and contrastive reasons: because the requisite structure provably holds globally, this connection delivers genuine formal guarantees and thereby opens the door to a broad class of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

This section provides the formal definitions of the models and distributions referenced throughout the paper: Subsection B.1 defines the model families, and Subsection B.2 formalizes the distributions.

B.1 MODEL FORMALIZATIONS

This subsection formalizes the three base-model types analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. We define a decision tree (DT) as a directed acyclic graph representing a function $f : \mathbb{F} \rightarrow [c]$ over $c \in \mathbb{N}$ classes, whose structure encodes the function as follows: (i) each internal node v is labeled with a distinct binary input feature from $\{1, \dots, n\}$; (ii) each internal node v has at most k outgoing edges, one per value in $[k]$ that may be assigned to v ; (iii) along any root-to-leaf path α , every variable appears at most once; (iv) each leaf is labeled with a class in $[c]$. Consequently, any input assignment $\mathbf{x} \in \mathbb{F}$ traces a unique root-to-leaf path α , and the output $f(\mathbf{x})$ is the class label $i \in [c]$ of that leaf. The size $|f|$ of the DT is the total number of edges in the graph. For modeling flexibility, distinct paths α and α' may test the input variables $\{1, \dots, n\}$ in different orders, subject only to the constraint that no variable is repeated along any single path.

Neural Networks. We state our *hardness* proofs for neural networks with ReLU activations, while our *tractability* results (polynomial-time algorithms) require only the standard assumption of polynomial-time inference. Consequently, every tractable-versus-intractable separation we prove transfers to *any* neural architecture at least as expressive as a standard feed-forward ReLU network — a class that subsumes the models most widely deployed in practice, including ResNets, CNNs, Transformers, and diffusion models. Since any ReLU network can be rendered fully connected by assigning zero weights and biases to absent connections, we follow standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolfi et al., 2025) and assume full connectivity without loss of generality; that is, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP f comprises $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) and one output layer $g^{(t)}$, with each layer defined recursively by:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$, $b^{(j)}$, and $\sigma^{(j)}$ denote respectively the weight matrix, bias vector, and activation function of the j -th layer. The model is therefore parameterized by t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, the binary input vector. The network’s dimensions are fixed by a sequence of positive integers $d_0, \dots, d_t$, with the (training-learned) parameters satisfying $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$. As the model is a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers employ the ReLU activation $\text{ReLU}(x) = \max(0, x)$. Although training typically uses a sigmoid output activation, for the purpose of post-hoc interpretation we take the output layer to apply the threshold-based step function $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Because our goal is post-hoc explanation, we characterize ensembles by their *inference* behavior rather than their training procedure. Our analysis targets the two dominant inference schemes: *majority voting* (e.g., random forests) and *weighted voting* (e.g., XGBoost). As with neural networks, however, our *tractability* results — the polynomial-time algorithms — require only polynomial-time inference and therefore extend to an even broader range of ensemble configurations.

Majority Voting Inference. Under *majority voting*, the prediction $f(\mathbf{x})$ is the class $j \in [c]$ receiving the plurality of votes among the individual tree predictions $f_i(\mathbf{x})$ over all $i \in [k]$:

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ returns the most frequent label in the multiset S , ties being broken by a fixed rule (e.g., lexicographic order or a predefined priority).

Weighted Voting Inference. Under *weighted voting*, each ensemble member carries a weight $\phi_i \in \mathbb{Q}$ encoding its relative importance, and the predicted class is the one accumulating the greatest total weight. Formally, for any $\mathbf{x} \in \mathbb{F}$:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the indicator function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection formalizes the distribution classes introduced in the main paper.

Empirical Distributions. An empirical distribution $\mathcal{D}$ over the input features is induced by a dataset $\mathbf{D}$ of inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$, assigning each $\mathbf{x} \in \mathbb{F}$ a probability equal to its relative frequency in $\mathbf{D}$:

$$\Pr(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Given a per-feature marginal $p(\mathbf{x}_i) \in [0, 1]$ for each input feature, the distribution $\mathcal{D}$ is *independent* when the joint probability factorizes as $\Pr(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

When $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ for all $i, j \in [n]$, this reduces to the *uniform distribution*, itself a special case of an independent distribution.

General Distributions. Many of our proofs — especially those establishing the fundamental properties of the value functions — hold under fully general distributional assumptions, and this generality broadens their applicability well beyond any single modeling choice. While empirical distributions (using the training set as a proxy) are the prevailing choice in XAI, the framework equally accommodates alternative distribution estimators, including k -NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for tabular dependencies (Größer & Okhrin, 2022), and expressive conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022) — estimators widely used in the counterfactual-explanation and algorithmic-recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024) to which contrastive explanations are closely related. When the value function is computed neither from a structural property of the model (e.g., leaf enumeration in decision trees) nor from an empirical distribution, it can instead be estimated by Monte Carlo sampling (Hastings, 1970), following approaches akin to (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

This subsection examines the distinction between cardinal and subset minimality in greater detail. Cardinal minimality is the substantially stronger guarantee, corresponding to globally minimal explanation size, whereas subset minimality guarantees only a local form of minimality.

That cardinal minimality implies subset minimality is immediate: if $S \subseteq [n]$ is cardinally minimal, no set S' with $|S'| < |S|$ is an explanation, so in particular no strict subset $S' \subsetneq S$ is an explanation, and hence S is subset-minimal.

The converse, however, fails. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

together with the assignment $\mathbf{x} := (1, 1, 1)$, so that $f(1, 1, 1) = 1$. Fixing $\mathbf{x}_1 = 1$ yields a cardinally minimal (and thus subset-minimal) sufficient reason, since the prediction stays 1 irrespective of $\mathbf{x}_2, \mathbf{x}_3$. Fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ also yields a subset-minimal explanation — yet not a cardinally minimal one, since it has larger cardinality. This confirms that every cardinally minimal explanation is subset-minimal, but not conversely.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024), a phenomenon exhibited by functions of the form below. We show that this same function admits an exponential number of *global* and *probabilistic* sufficient reasons, thereby transferring the exponential blow-up to our setting. The function is a special case of the broader class of *threshold Boolean functions* of Wegener (2005): for $n = 2k + 1$ with $k \in \mathbb{N}$, it is defined by:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. The function f is *symmetric* in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends only on the number of 1's (equivalently, 0's) in the input and is invariant under every permutation of the input bits. Moreover, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Because f is symmetric, the equality $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$ of equal size. Furthermore, one verifies that for any subset $S \subseteq [n]$ of size $k + 1$:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and set $k := \lfloor \frac{n}{2} \rfloor$. Each such subset is *subset-minimal*, since every subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ at the threshold δ . The counting argument of (Bassan et al., 2024) then applies verbatim, yielding exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. By Stirling's approximation:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This establishes the claimed $\Theta(\frac{2^n}{\sqrt{n}})$ bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. We establish monotonicity for both families of global explanations — sufficient and contrastive. We begin with global sufficiency, first treating the illustrative Boolean case to build intuition, then generalizing successively to functions with discrete multi-valued input domains and multiple output classes, and finally to continuous input domains, thereby covering every well-defined classification function. We then derive the corresponding guarantee for global contrastive explanations. Crucially,

none of these proofs invokes feature independence: monotonicity holds in the strongest possible sense, for *any* underlying distribution.

We close by exhibiting a simple example demonstrating that monotonicity fails for the local probabilistic sufficient and contrastive value functions, confirming that the property is genuinely a global phenomenon.

Lemma 1. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

For notational economy, we occasionally suppress the $\sim \mathcal{D}$ subscript in the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and write f^+ and f^- for the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We first carry out a technical simplification of the probability term, which streamlines the subsequent monotonicity argument.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to S . Fixing a point $\mathbf{x} \sim \mathcal{D}$ and a set S , all points $\mathbf{z} \sim \mathcal{D}$ with $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, for $b \in \{0, 1\}$, share the identical local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. Marginalizing over $\mathbf{x}_{\bar{S}}$ across all $\mathbf{x} \sim \mathcal{D}$ that agree on $\mathbf{x}_S$ and produce the same output b gives $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, whence:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. Building on the preceding simplification, we now prove the monotonicity claim, first introducing some additional notation. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$, and analogously $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\
&\geq 0.
\end{aligned} \tag{26}$$

Here we have used the identities $P_1 - P_1^2 = P_1(1 - P_1) = P_1P_0$ and, symmetrically, $P_0 - P_0^2 = P_0(1 - P_0) = P_0P_1$. Together with the preceding derivation, these establish that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, completing the proof. $\square$

Lemma 2. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. We now lift the Boolean argument to functions whose input domains and output ranges are each finite sets of values, letting $\mathcal{K}$ denote the set of output classes. Equation 22 generalizes to:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \\
&\sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right]
\end{aligned} \tag{27}$$

Conditioning on $\mathbf{z}_S = \mathbf{x}_S$ and letting feature t range over its r possible values $\{v_1, \dots, v_r\}$, equation 23 generalizes to:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right].
\end{aligned} \tag{28}$$

Let P_k denote $\Pr(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Therefore $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, completing the proof for the discrete multi-valued setting. $\square$

Lemma 3. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend monotonicity to functions with continuous feature domains. One route partitions each continuous feature's domain into two disjoint subdomains and adapts the proof of lemma 2 by adjusting notation only; since both summation and integration are linear, the core reasoning carries over unchanged from the discrete to the continuous case.

For completeness, we instead give a direct proof. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the joint probability density, $p(x_i)$ the marginal density of x_i , and $\mathcal{X}_i$ the domain of feature i .

We define the marginal density by

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where $\mathcal{X}_{\bar{S}}$ is the domain of the features in $\bar{S}$, and the conditional density by:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of S given $\mathbf{x}$ is then

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

which decomposes, via an additional feature $t \notin S$, as:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

Equation 27 accordingly becomes

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 becomes

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For brevity, write $Y_j(t)$ for $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$ and $p_{t|S}$ for $p(z_t \mid \mathbf{x}_S)$; both are implicitly associated with a fixed $\mathbf{x}_S$ ². Monotonicity then follows from:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. Combining lemmas 1 to 3, the global sufficient value function v_{suff}^g is monotonic non-decreasing for every well-defined classification function — Boolean, discrete, continuous, or any combination thereof — under any data distribution.

$$v_{con}^g(S) = 1 - v_{suff}^g(\bar{S}) \quad (38)$$

Hence v_{con}^g is monotonic non-increasing as features are added to $\bar{S}$ or, equivalently, monotonic non-decreasing as features are removed from $\bar{S}$ and added to S , which completes the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This section establishes supermodularity for both global sufficient and contrastive explanations. As with monotonicity, we proceed from Boolean functions to multi-valued discrete and then continuous input domains, ultimately covering every well-defined classification function. In contrast

²Changing $\mathbf{x}_S$ alters their values; we suppress $\mathbf{x}_S$ in the notation for brevity.

to monotonicity, however, these results require the assumption of feature independence, and we substantiate this requirement with a counterexample showing that supermodularity can indeed fail once independence is relaxed.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Concretely, we prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

for $S \subseteq S'$ and $t \notin S'$ — i.e., that the marginal gain $\Delta_t(S, f)$ is itself monotone in S .

The case $S' = S$ is immediate. It thus suffices to treat $S' = S \cup \{i\}$ with $i \notin S$ and $i \neq t$. Under feature independence, the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$ decomposes as:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) &= \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ &\Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ &\Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ &\Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Write P_1, P_0, Q_1 , and Q_0 for $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively, and let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This implies that $\Delta_t(S, f)$ is monotone, that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote $\Pr(z_i = v_{k'})$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$. We decompose equation 30 to

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right]. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_i = v'_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].
\end{aligned} \tag{45}$$

Therefore, we get

$$\begin{aligned}
&\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{46}$$

□

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that $\mathcal{D}_p$ satisfies feature independence. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we provide a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \tag{47}$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \tag{48}$$

We let $Y_j(i, t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
&\Delta_t(S, f) \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\ &= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}. \end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned} & \Delta_t(S', f) \\ &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \quad (50) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S, \end{aligned}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\ &= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}. \end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned} & \Delta_t(S', f) - \Delta_t(S, f) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (51)$$

Consequently, the global sufficient value function v_{suff}^g retains the supermodularity property for any valid classification function, regardless of whether its features are Boolean, discrete, continuous, or a mix thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \quad (52)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , we have that the following holds:

$$\begin{aligned} v_{\text{suff}}^l(\emptyset) &= \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4}, \\ v_{\text{suff}}^l(\{1, 2\}) &= \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1. \end{aligned}$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. □

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once we establish that the global sufficient setting is supermodular, it directly follows that the global contrastive setting is submodular. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. To compare the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently compare $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, hence concluding this proof.

We will now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$\begin{aligned} v_{\text{con}}^l(\emptyset) &= 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2}, \\ v_{\text{con}}^l(\{1, 2\}) &= \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}. \end{aligned}$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. □

In the rest of this section, we show that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Moreover, define a distribution $\mathcal{D}$ on x_1 and x_2 . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, we have that the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) \\
& = (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2
\end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1)
\end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)}
\end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1)
\end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0)} + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1)} + \\
& \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 1, x_2 = 0)} + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= 1
\end{aligned}$$

Let $t = 2$. Evidently, we have $\Delta_t(\{1\}, f) = 0$. Next, let us analyse $\Delta_t(\emptyset, f)$. We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2.
\end{aligned}$$

This implies $\Delta_t(\emptyset, f) = 0$, that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. However, without the assumption of feature independence, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. For example, let:

$$\mathbf{Pr}(x_1 = 0, x_2 = 0) = P_{00}, \mathbf{Pr}(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\mathbf{Pr}(x_1 = 1, x_2 = 0) = P_{10}, \mathbf{Pr}(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. We note that this proposition follows directly from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (as established in Proposition 2), along with other relevant conclusions drawn in prior work. In the remainder of this section, we elaborate on how this monotonicity property leads to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g follows directly from the monotonicity property, for which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) showed that this type of greedy algorithm yields a subset-minimal explanation when the underlying value function is monotone non-decreasing. The algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ drops below δ , ensuring that although $v(S) \geq \delta$ (a maintained invariant), removing any element causes the condition to fail. Due to monotonicity, this implies that no proper subset of S satisfies the condition, guaranteeing that S is subset-minimal.

We now turn to explain why Algorithm 1 does not converge to a subset-minimal solution in the local setting. We note that it is well known that the local probabilistic sufficient value function (and likewise the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables $\{x_1, x_2\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$ which is classified as 1, then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , the S denotes the set of features that are allowed to vary.) Because $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is also non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We begin by noting that for decision trees, assuming that $\mathcal{D}$ represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

Can be carried out in polynomial time using the following procedure. Thanks to the tractability of decision trees, we can iterate over all pairs of leaf nodes, each corresponding to partial assignments $\mathbf{x}_S$ and $\mathbf{z}_S$. For each such pair, if both leaf nodes yield the same prediction under f , we compute the corresponding term in the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, allowing the full expectation to be computed by summing the contributions of all such matching leaf pairs.

Since all probabilities involved are provided as part of the input and each step involves only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step in Algorithm 1 is efficient, and thus the algorithm as a whole runs in polynomial time. Finally, combining this with Lemma 2, which proves that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation. This is because the algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , while $v(S) \geq \delta$ is preserved throughout. By monotonicity, this guarantees that no strict subset of S satisfies the condition, ensuring S is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming $P \neq NP$, there is no polynomial-time algorithm for computing a subset-minimal δ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

Can be performed in polynomial time when $\mathcal{D}$ is selected from the class of empirical distributions. This is achieved by iterating over pairs of instances $\mathbf{x}, \mathbf{z}$ within the dataset and running an inference through $f(\mathbf{x})$ and $f(\mathbf{z})$ to compute the expected values at each step. Consequently, determining whether the probability function exceeds or falls below a given threshold δ can also be accomplished in polynomial time. Furthermore, leveraging the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we can utilize algorithm 1 for contrastive reason search or for sufficient reason search. By executing a linear number of polynomial queries, a subset-minimal explanation (whether a subset minimal contrastive reason or a subset minimal sufficient reason) can thus be obtained in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part of the proof follows directly from Proposition 6, which established that this holds for any model, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ can be computed in polynomial time from the empirical distribution. For the second part, we proceed by proving two lemmas: one establishing the intractability of computing the sufficient case, and the other addressing the contrastive case.

Lemma 6. *Unless $PTIME=NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local δ sufficient reason for neural networks, then it would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

Lemma 7. Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.

Proof. We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution $\mathcal{D}$ consists of a single element, which, for simplicity, we take to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to $\mathbf{0}_n$, we define $\mathbf{1}_n$ as an n -dimensional vector consisting entirely of ones. Given an input ϕ , we initially assign $\mathbf{1}_n$ to ϕ , effectively setting all variables to True. If ϕ evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

ϕ' , while no longer a CNF, can still be transformed into an MLP f using Lemma 8, ensuring that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Given our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we can determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, noting that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We now assert that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Let us assume that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is $\emptyset$ in this case, this trivially holds. The fact that S is subset-minimal means that any other subset $S' \subseteq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution contains only the point $\mathbf{0}_n$), the probability function can only take values of 1 or 0. Hence, we also know that $v(S') = 0$. This tells us that, aside from the subset $S = \{1, \dots, n\}$, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the rest to 0 does not result in a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly implies that ϕ is assigned False for any of these inputs. Since we already know that ϕ does not return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now, suppose that S is a subset that is strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we can apply the same reasoning as before to conclude that $v(S) = 1$. This implies that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False. Consequently, the ϕ component must be True, which implies that $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We observe that we can apply the same process used in our proof for neural networks, where we encoded ϕ' into an equivalent neural network. However, ϕ' is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since ϕ' consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. *Unless $PTIME = NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of ϕ .

First, we check whether assigning all variables in ϕ to 1 evaluates ϕ to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network f . Next, we compute a subset-minimal δ -sufficient reason by setting $\mathcal{D}$ as the empirical distribution containing only a single data point $\mathbf{0}_n$, following a similar procedure to Lemma 6. Additionally, we set $\delta = 1$ and compute a subset-minimal δ -contrastive reason concerning $\langle f, \mathbf{x} \rangle$.

We will now demonstrate that if any subset-minimal δ contrastive reason obtained for ϕ is satisfiable, then ϕ itself is satisfiable. Conversely, if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$, where the distribution $\mathcal{D}$ considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a $\delta = 1$ contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of f , which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates f (and consequently ϕ) to True, implying that ϕ is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature*

independence. In contrast, unless $PTIME=NP$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

Proof. We divide the proof of the theorem into two lemmas, covering both the approximation guarantee and the result on the absence of a bounded approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We will in fact prove a stronger claim, showing that this holds for *any* model, provided the trivial condition that its inference time is computable in polynomial time, along with one additional mild condition that we will detail later — both of which apply to both our neural network and tree ensemble formalizations.

We begin by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions v with integer values. More generally, their result yields an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that the value function v_{con}^g is monotone non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must also show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|D|))$, implying that it is effectively constant, due to the assumption of a fixed dataset $|D|$. To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to be zero. We begin by formally defining what we mean by redundancy:

Definition 1. *Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.*

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ indicates that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ indicates that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, in this sense, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then for any input of size n' , the output of f' will exactly match the output of f when applied to the same features $[n] \setminus \{i\}$. This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles, it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as previously discussed in this proof and in Proposition 2 — we can compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in D$ and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature i is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then i is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution $\mathcal{D}$, we can remove them and construct an equivalent model f' . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Therefore, once the preprocessing step removes all redundant features from f and we construct the resulting function f' , we ensure that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|\mathcal{D}|))$. Given that the empirical distribution $\mathcal{D}$ is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|\mathcal{D}|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|\mathcal{D}|^2}$. In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

Which concludes our proof.

□

Lemma 11. *Unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.*

Proof. Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble—even when $|D| = 1$. However, Lemma 9 establishes that even with a single baseline $\mathbf{z} = 0_n$ (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k_f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We divide the proof into two lemmas: one establishing the approximation guarantee for the global case, and the other demonstrating the intractability of the local case.

Lemma 12. Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.

Proof. The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of $\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)$. After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. We established in Proposition 2 that the value function v_{suff}^g is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from f — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that behaves identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from f . We note that here too, it holds that a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. Hence, this preprocessing phase will give us a new function f' for which it both holds that it is equivalent to the behaviour of f for all remaining features and also satisfies that for any i it holds that both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by $\frac{1}{|D|}$ and its expected value by $\frac{1}{|D|^2}$, we obtain that $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. This implies, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|D|))$, and for the second term we derive the following:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$. This implies: $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, and therefore, $1 - |D|^2 \leq k^f \leq 0$. This yields $\frac{1}{1-k^f} \leq |D|^2$, demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

Lemma 13. *Unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble—even when $|D| = 1$. Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline $\mathbf{z} = 0_n$, deciding whether a sufficient reason exists is NP-hard unless $PTIME = NP$. Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $NP \subseteq DTIME(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ defined over a dataset $\mathbf{D}$, and some threshold δ' . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f . Before doing so, we specify the functional form that we intend the constructed neural network f to implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, such that they agree on all coordinates outside S_j ; that is, they share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = \mathbf{y}^j \quad (76)$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — that is, no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total, this yields $2m$ vectors that form the dataset $\mathbf{D}$ from which we construct f .

We then construct a neural network f that simulates this function. This is achieved using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network capable of representing any finite set of input–output pairs can be built in polynomial time. Using this result, we construct f accordingly, and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

We now argue that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair of vectors $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term. Moreover, because each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, there is no overlap between these pairs. Consequently, the coverage achieved by any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We begin by defining orthogonal DNFs (Crama & Hammer, 2011) before presenting the full proof.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

This means that no single variable assignment can satisfy more than one term simultaneously.

The proof mirrors the argument used in Theorem 1, relying on the fact that v_{suff}^g can be computed in polynomial time for any set S .

2214 For a given set S , let $\mathcal{C}^S T_i$ be the constraints that the term T_i imposes on the variables in S .
 2215 We enumerate all pairs of terms (T_i, T_j) . Any pair whose constraints on S are *inconsistent* (i.e.,
 2216 $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$) is ignored, since it contributes zero to v^g_{suff} .
 2217

2218 For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$.
 2219 The contribution of this pair to v^g_{suff} is then
 2220

$$2221 \Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

2222
 2223 Under feature independence, these probabilities factor as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Since there are at
 2224 most m^2 such term pairs, the value of v^g_{suff} can be computed in polynomial time under an independent
 2225 distribution $\mathcal{D}$. $\square$
 2226

2227 O DISCLOSURE: USAGE OF LLMs

2228
 2229 An LLM was used exclusively as a writing assistant to refine grammar and typos and improve clarity.
 2230 It did not contribute to the generation of research ideas, study design, data analysis, or interpretation
 2231 of results, all of which were carried out solely by the authors.
 2232
 2233
 2234
 2235
 2236
 2237
 2238
 2239
 2240
 2241
 2242
 2243
 2244
 2245
 2246
 2247
 2248
 2249
 2250
 2251
 2252
 2253
 2254
 2255
 2256
 2257
 2258
 2259
 2260
 2261
 2262
 2263
 2264
 2265
 2266
 2267